Lab Work: How to build a virtual microscope

The main objective of this lab work is to understand the principles behind image formation in an optical microscope through simulation, in particular how the parameters of the experimental setup affect the final image resolution. Students will program in MATLAB a complete imaging pipeline: from object definition, through light-sample interactions and pupil filtering by the objective, to final detection on a camera sensor. The goal is to develop, step by step, a complete code for an imaging simulation through a microscope.

Learning Objectives

- Understanding the concept of resolution and the role of the numerical aperture during pupil filtering.
- Learning how different imaging modes (brightfield, darkfield, phase contrast) can be modeled.
- Simulating the effect of finite detector sampling and noise on the image formation.
- Exploring defocus and depth imaging using Fresnel propagation.
- Designing a complete virtual microscope with realistic off-the-shelf components.

Instructions

This lab work is structured as a sequence of tasks. Each task introduces a concept and requires you to implement corresponding MATLAB code. You are expected to produce functions and scripts that can be assembled into a single pipeline at the end of this session to be handed over to the professor in charge. Your work will be assessed and graded both on the code you provide and the comments and explanations alongside it.

Deliverables

At the end of the session, you should have:

- A set of MATLAB functions (makeObject, computePupil, applyPupilImaging, etc).
- A main script that calls these functions in sequence.
- Figures illustrating your simulation.

0. Overview

In the form of a diagram, summarize the imaging pipeline through the microscope and the different functions needed for your simulation.

1. Defining the Object

We begin with a synthetic biological object, representing (mostly) transparent white blood cells (about 10) with dark nuclei in them.

Q1. Write a MATLAB function makeObject that creates both a phase map and an intensity map of the object.

Use the following parameters:

```
n_cells = 10;
cell_radius_mean = 50e-6;
cell_radius_sigma = 10e-6;
nucleus_radius_frac = 0.10;
delta_n = 0.02;
lambda_ref = 550e-9
```

Q1.1 What would the image of such an object look like on a traditional widefield microscope? What is the minimum numerical aperture that should allow you to resolve the nuclei?

Q1.2 What kind of technique could be used to retrieve information from a sample that is mostly transparent?

2. Resolution and the Pupil Function

The imaging system is modeled by its pupil function.

Q2. Implement computePupil that returns a binary pupil mask of the adequate shape and size.

Q3. Using fft2, fftshift, ifft2, simulate imaging of the object for NA = 0.01, 0.1, 1.0. Compare the resulting images. Are the cells resolved? The nuclei?

Q3.1 Where is the pupil of the objective physically located?

Q3.2 Intuitively, how come the process of imaging through an optical system involves Fourier transforms?

Q3.3 A target can be used to quantify the resolution of a microscope. Use the different targets provided to try and assess the resolution of your system. Explain the method you are are using as well as your results.

3. Imaging Contrast Mechanisms

Q4. Set NA=0.5. Modify the pupil to simulate darkfield imaging (annular pupil). Compare the image to brightfield.

Q5. Implement a weak object transfer function (WOTF) by linearizing the phase (field ≈ 1 + i*φ) to simulate a phase contrast microscope. Display the resulting phase-contrast image.

4. Detector Sampling and Noise

Q6. Write a function addCameraEffects that:

- Resamples the image according to pixel pitch (use imresize).
- Adds Poisson noise (poissrnd).

Simulate the same object with two cameras: Pixel pitch 1 μm, SNR=30 and 3.5 μm, SNR=60 and explain how both parameters affect your ability to resolve the objects.

5. Defocus and Depth of Field

Q7. Implement a fresnelPropagate function using the angular spectrum method to simulate a 20 μm object defocus.

Q8. Extend the simulation to a two-layer object (two phase maps separated by dz). Explain your reasoning behind how you constructed this simulation. Show how both layers contribute to the image.

6. Assembling a Complete Microscope

Q9. Write an assembleMicroscope function that:

- Applies propagation and pupil filtering.
- Simulates the detector response with noise.

Q10 Visit Thorlabs' website and find what seems to be the most appropriate component for your setup. Test the complete function with realistic parameters and produce a final simulated image.

Q11 Using the provided real image of cells, simulate the image you should obtain through your microscope.